

Predicting Home Price Direction Using Historical Housing and Macroeconomic Data

White Paper
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Business Problem & Research Questions

For this project, I wanted to look at whether housing prices are likely to go up or down based on economic factors such as mortgage rates, unemployment, and inflation. I focused on the 15 largest U.S. metropolitan areas and looked at whether these factors relate to home price changes differently across each one. Specifically, I'm asking: can mortgage rates, unemployment, and inflation predict whether home prices in a given metro area will rise or fall the following month, and do these factors matter more in some metro areas than others? Homebuyers, investors, and lenders all want to know where the housing market is headed before making big financial decisions, so this project explores whether economic indicators can actually help answer that.

Background / History

Housing affordability and price swings have become a bigger concern for households and policymakers in recent years. The 2008 financial crisis showed how closely home values are tied to broader economic conditions, especially mortgage lending and interest rates. More recently, home prices jumped sharply from 2020 to 2022, then cooled off as mortgage rates climbed. That kind of volatility is part of what made me want to look at whether economic indicators can actually explain or predict these shifts. For buyers, sellers, and lenders, knowing whether a market is trending up or down has real financial consequences, which is what makes this worth studying.

Data Explanation

I combined two data sources for this project. The Federal Reserve Economic Data (FRED) API gave me monthly mortgage rate, unemployment rate, and inflation (CPI) data at the national level. Zillow's Home Value Index (ZHVI) gave me monthly home value data for the 15 metro areas I'm focusing on.

Getting the data ready took a few steps. FRED's mortgage rate data comes in weekly, so I averaged it into monthly values to match the unemployment and inflation data. FRED also uses "." as a placeholder for missing values, so I converted those to null before doing any calculations. On the Zillow side, I filtered the data down to my 15 target metros and reshaped it from wide format, with one column per month, into long format, with one row per metro per month, so it would merge properly. The two sources didn't share the same date format either. Zillow uses month-end dates and FRED uses month-start dates, so I standardized both to a year-month format before joining them.

I also built a few features to add more time-based context: one-month lagged values for each economic indicator, the month-over-month percent change in home value, and a 3-month rolling average of home value. My target variable, `price_direction`, is a binary flag for whether a metro's home value went up the following month. A handful of rows ended up with missing values because of the lag and rolling calculations, mostly the first month or two for each metro, so I dropped those before modeling. The full data dictionary is in Appendix A.

Methods

I'm treating this as a binary classification problem: predicting whether a metro's home value will go up or down the following month, rather than trying to predict an exact price. Classification made more sense for this project since it gives a clear, useful signal for buyers, investors, and lenders without overstating how precise the prediction actually is.

I started with logistic regression as a baseline model since it's simple and easy to interpret. Right away I ran into a problem: about 75% of the months in my data showed prices going up, so the model just defaulted to predicting "up" almost every time instead of actually learning the pattern. To fix this, I scaled the features and used class weighting, which helped but still left the model weak at correctly identifying months where prices went down (F1 score of 0.53).

From there I moved to Random Forest as my more advanced model, since it can pick up on more complex relationships in the data and handles class imbalance better than logistic regression. Random Forest performed noticeably better across every metric, so I used it as my primary model going forward.

For validation, I used a time-based train and test split instead of random cross-validation, training on data through December 2021 and testing on everything from January 2022 onward. This approach, sometimes called walk-forward validation, avoids letting the model "see" future data during training, which would give a misleadingly good result.

Analysis

To validate the models, I used accuracy, F1 score, ROC-AUC, and confusion matrices, since these give a fuller picture than accuracy alone, especially with the class imbalance I ran into.

My tuned logistic regression model, using a 0.4 probability threshold, reached 59% accuracy and an F1 score of 0.53. Its ROC-AUC was actually quite strong at 0.94, which told me the model's underlying probability scores were good at separating "up" from "down" months, even though its default classification threshold wasn't doing a great job translating that into correct predictions.

Random Forest performed much better across the board: 88% accuracy, an F1 score of 0.91, and a similar ROC-AUC of 0.94. Looking at the confusion matrix (Figure 2), Random Forest correctly identified 233 out of 299 "down" months and 469 out of 496 "up" months in the test set. It still made more mistakes labeling actual price decreases as increases (66 cases) than the reverse (27 cases), but overall it handled both classes far better than logistic regression did (Figure 4).

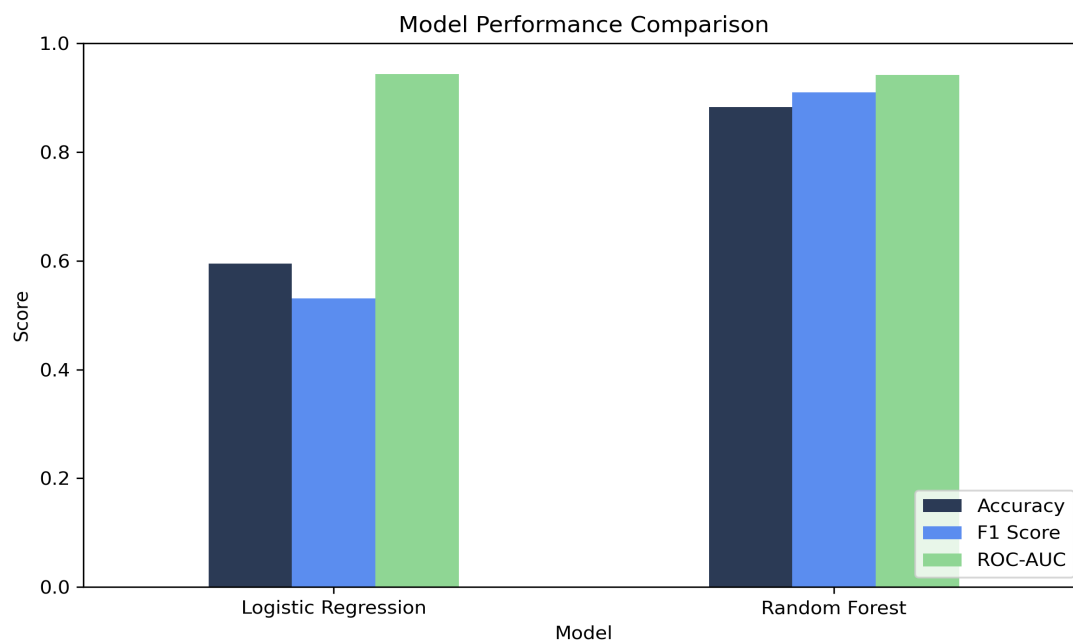


Figure 1. Accuracy, F1 score, and ROC-AUC comparison between logistic regression and Random Forest.

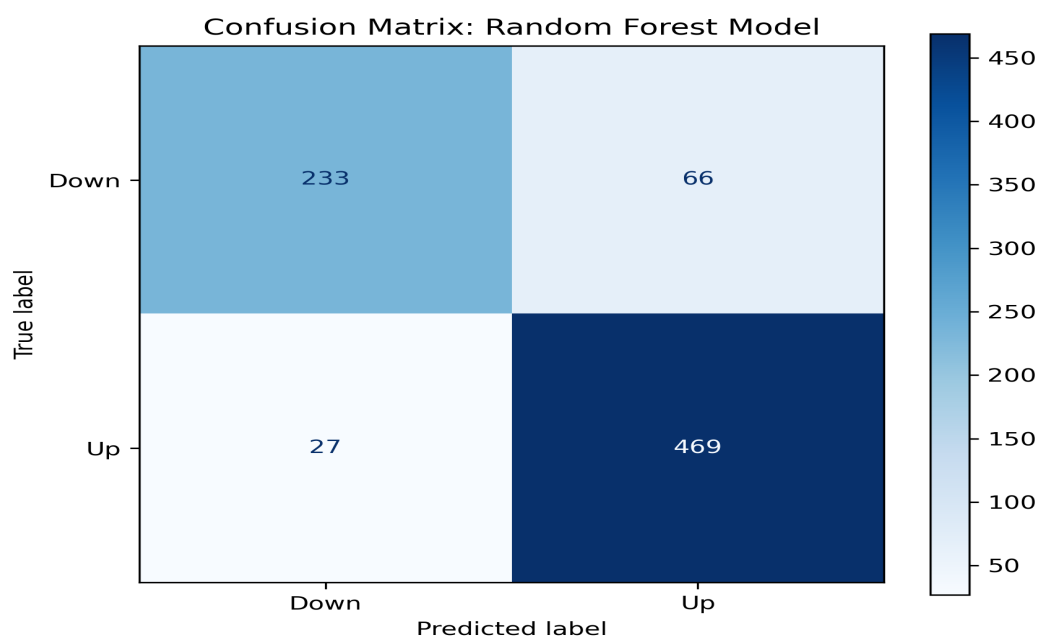


Figure 2. Confusion matrix for the Random Forest model on the test set.

I also looked at feature importance from the Random Forest model to see which factors mattered most (Figure 3). Month-over-month home value change was by far the strongest predictor, accounting for about 46% of the model's decision-making, followed by CPI and lagged CPI (about 30% combined). Unemployment and mortgage rate mattered less, likely because they tend to move more gradually month to month, while recent price momentum and inflation shifts more

directly signal what's about to happen next.

So, to answer my original research questions: yes, mortgage rates, unemployment, and inflation, along with recent price momentum, can meaningfully predict home price direction, particularly when using a model like Random Forest that can capture non-linear relationships. Recent price momentum and inflation carried the most predictive weight overall, while mortgage rates played a smaller role than I initially expected going into this project.

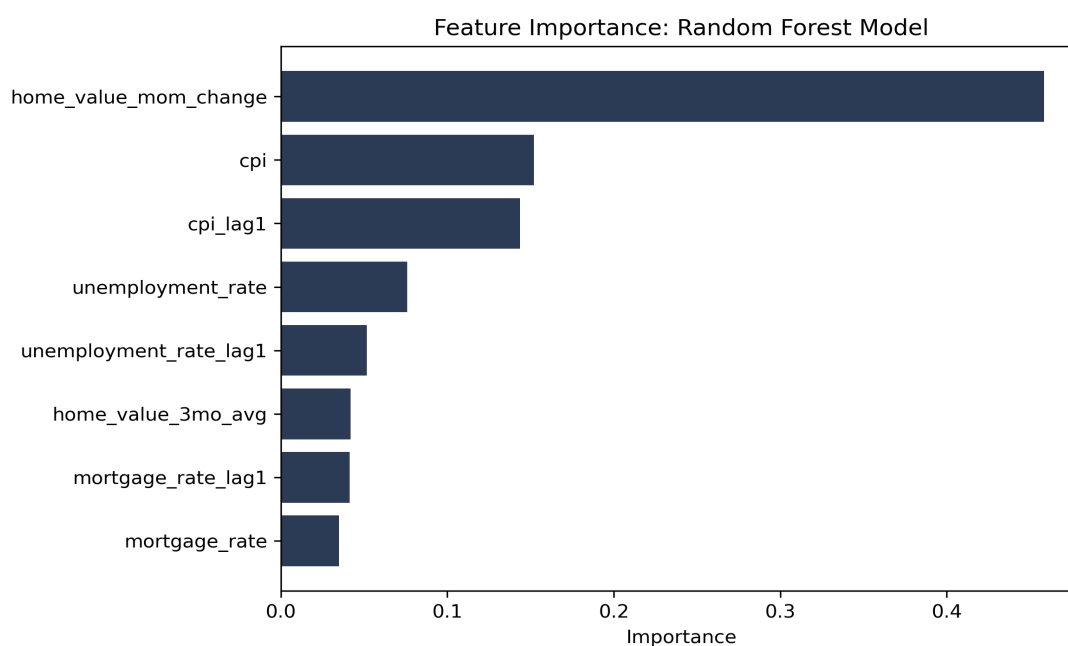


Figure 3. Feature importance from the Random Forest model.

Conclusion

This project set out to answer whether macroeconomic and housing market indicators can predict home price direction across major U.S. metro areas, and the results say yes, with some important caveats. Random Forest correctly predicted price direction 88% of the time on data it hadn't seen before, and the strongest signals turned out to be recent price momentum and inflation, not mortgage rates like I originally expected going in.

For a domain audience like buyers, investors, or lenders, the key takeaway is that recent trends and inflation conditions are more useful short-term signals than mortgage rate changes alone. That said, this model is best used as one input among several, not a standalone prediction tool, especially since it leans heavily on recent momentum, which could make it less reliable during sudden market shifts.

Assumptions

A few assumptions shaped this project. I assumed that national-level economic indicators, like mortgage rates, unemployment, and inflation, are reasonably representative conditions for all 15 metro areas, even though local economic conditions can vary. I also assumed that Zillow's

smoothed, seasonally adjusted ZHVI accurately reflects real home value trends, since I didn't have access to raw, unadjusted sale price data to compare it against. Finally, I assumed that a one-month prediction window is a meaningful and useful timeframe for the audience this project is meant to help, rather than, say, a 3 or 6-month outlook.

Limitations

There are a few limitations worth being upfront about. The model relies heavily on recent price momentum, which means it may perform worse during sudden market shifts or economic shocks, like 2008 or 2020, since those kinds of events don't follow the typical patterns the model learned from historical data. Using national-level economic indicators instead of metro-specific ones is another limitation, since it assumes mortgage rates, unemployment, and inflation affect all 15 metro areas similarly, when in reality local conditions can vary quite a bit.

The model also still misclassifies actual price decreases as increases more often than the reverse, so it tends to be more reliable at catching upward trends than downward ones. Lastly, this project only predicts direction, up or down, not the size of the change, so it can't tell you whether a "down" month means a small dip or a larger drop.

Challenges

The biggest challenge was dealing with class imbalance in the target variable, since about 75% of months in the data showed price increases. My first model defaulted to predicting "up" almost every time, which looked fine on accuracy alone but was actually a weak model. Fixing this took some trial and error, including scaling the features, adjusting class weights, and testing different probability thresholds before moving to a different model type altogether.

Aligning the FRED and Zillow datasets was another challenge, since they used different date formats and different reporting frequencies, weekly for mortgage rates versus monthly for the other indicators. I had to standardize both to a shared year-month format and average the weekly mortgage data into monthly values before I could merge everything together correctly.

Future Uses / Additional Applications

This approach could be extended in a few directions. Adding metro-specific economic indicators, like local unemployment rates or regional building permit data, instead of relying only on national figures, could improve accuracy and better capture how different markets respond to the same conditions. The model could also be adapted to predict a range of price movement instead of just direction, giving users a clearer sense of how big a change might be, not just which way it's headed.

This same framework could also be applied beyond home prices, for example predicting rental price trends, or extended to a longer prediction window, like a 3 or 6-month outlook, for users who need to plan further ahead than one month.

Recommendations

Based on these findings, I'd recommend that buyers, investors, and lenders pay closer attention to recent price momentum and inflation trends when trying to gauge where a housing market is headed, since these mattered more in this model than mortgage rate changes did. That said, this model should be used as one input among several, not a standalone decision-making tool, especially given its tendency to be more reliable on upward trends than downward ones.

For anyone using this in practice, I'd also recommend focusing more attention on metro areas with more volatile price momentum, since the model's confidence and accuracy may vary more in those cases compared to steadier markets.

Implementation Plan

To put this into practice, the model would need to be re-run monthly as new FRED and Zillow data becomes available, since both are updated on a monthly basis. This means setting up a simple pipeline to automatically pull the latest FRED data through the API, download the newest Zillow ZHVI file, and re-run the same cleaning, feature engineering, and prediction steps used in this project.

For a real-world tool, predictions could be presented as a simple dashboard or report showing each metro area's predicted direction for the upcoming month, along with the confidence level behind that prediction, so users could weigh how much to rely on it. Before using this in any actual financial decision-making, the model would need ongoing monitoring to make sure its accuracy holds up over time, especially since it leans heavily on recent price momentum, which could break down during unusual market conditions.

Ethical Assessment

Identifying the Concerns

A few ethical concerns are relevant to this project. The biggest one is bias: since the model relies on national-level economic indicators rather than metro-specific ones, it could underrepresent local conditions in a way that unintentionally disadvantages certain markets or communities if used for real decisions like lending. There's also a fairness concern tied to the model's tendency to misclassify price decreases more often than increases, which means predictions may be more reliable for some markets or situations than others. Finally, there's a potential for misuse: if someone treated this model's predictions as guaranteed outcomes rather than one input among many, it could lead to poor financial decisions, especially in housing, where the stakes for individuals and families are high.

Assessing Likelihood and Impact

The risk of national-level indicators masking local conditions is fairly likely, since it's a structural limitation of the data I used, and the impact could be moderate to high if this were used to inform lending or investment decisions without additional local context. The risk of misclassifying downward trends is also fairly likely, given what the confusion matrix showed, and its impact would mainly affect users relying on the model during a market downturn, potentially leading them to

expect continued growth when prices are actually falling. The risk of misuse or overreliance on the model's predictions is harder to predict in terms of likelihood, since it depends on how the tool is actually used, but the potential impact is high, since housing decisions involve significant financial risk for individuals.

Mitigation

To address the risk of masking local conditions, I'd recommend incorporating metro-specific economic data in any future version of this model rather than relying solely on national indicators, and being transparent about this limitation whenever predictions are shared. For the imbalance in prediction reliability between upward and downward trends, I addressed this directly during modeling by using class weighting, which improved but didn't fully resolve the issue, so I've documented this limitation clearly rather than overstating the model's reliability. To reduce the risk of misuse, any real-world implementation of this model should be clearly labeled as a decision-support tool, not a guarantee, and paired with guidance encouraging users to consult additional sources before making financial decisions.

Data Changes

The main changes made to the data were reshaping Zillow's data from wide to long format, converting FRED's weekly mortgage data into monthly averages, and standardizing both datasets to a shared year-month format so they could be merged. Missing values represented by FRED's "." placeholder were converted to null and excluded from calculations rather than being filled in or estimated.

Legal or Regulatory Guidelines

Both data sources are publicly available and explicitly intended for research and analysis use, so no legal or regulatory restrictions applied to their use in this project. That said, if this model were adapted for actual lending or housing decisions in the future, it would need to be evaluated against fair lending laws, such as the Equal Credit Opportunity Act, given the model's reliance on economic indicators that can correlate with geographic and demographic patterns.

Risks Created by Transformations or How Results Are Presented

Averaging weekly mortgage data into monthly values smooths out some short-term volatility, which could obscure meaningful within-month rate changes. Presenting results as a simple "up or down" prediction also risks oversimplifying what is actually a probability-based estimate, so it's important that any presentation of results includes the model's confidence level and known limitations, rather than presenting predictions as certain outcomes.

Assumptions or Data Filtered Without Clearly Labeling It

Rows with missing values created by lag and rolling average calculations, typically the first one to three months of data per metro, were dropped before modeling. This was a necessary step for the model to function, but it does mean a small portion of early data for each metro wasn't included in training or testing.

Data Sourcing and Credibility

Both data sources used in this project, FRED and Zillow, are well-established and widely cited in economic and housing research. FRED is maintained by the Federal Reserve Bank of St. Louis, and Zillow's ZHVI is one of the most commonly referenced home value indices in the industry. Both datasets were accessed through their official public channels, FRED's API and Zillow's research data page, so the data was sourced and acquired ethically, with no scraping or unauthorized access involved.

Supporting Illustration

Figure 4 below illustrates why this research question matters in practice: home value trends vary significantly across metro areas, both in overall growth and in volatility during major economic events such as the 2008 financial crisis and the 2020-2022 pandemic-era price surge.

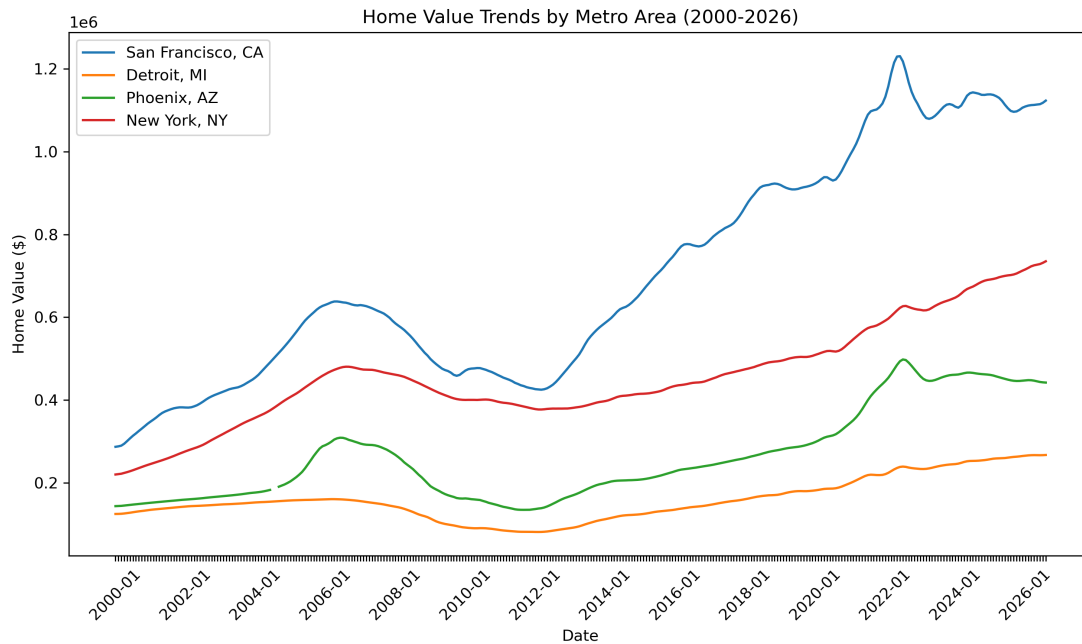


Figure 4. Home value trends for four contrasting metro areas, 2000-2026.

Appendix A: Data Dictionary

Field	Description	Source	Type
RegionName	Metro area name	Zillow ZHVI	Categorical
date	Year-month of observation	Zillow / FRED (aligned)	Datetime (YYYY-MM)
home_value	Zillow Home Value Index for the metro that month	Zillow ZHVI	Numeric (\$)
mortgage_rate	Avg. 30-yr fixed mortgage rate (national)	FRED (MORTGAGE30US)	Numeric (%)
unemployment_rate	National unemployment rate	FRED (UNRATE)	Numeric (%)
cpi	Consumer Price Index (inflation)	FRED (CPIAUCSL)	Numeric (index)
next_month_value	Home value the following month	Derived	Numeric (\$)
price_direction	Target: 1 = price increased, 0 = decreased	Derived	Binary (0/1)
mortgage_rate_lag1	Mortgage rate, previous month	Derived	Numeric (%)
unemployment_rate_lag1	Unemployment rate, previous month	Derived	Numeric (%)
cpi_lag1	CPI, previous month	Derived	Numeric (index)
home_value_mom_change	Month-over-month % change in home value	Derived	Numeric (%)
home_value_3mo_avg	3-month rolling average of home value	Derived	Numeric (\$)

References

Federal Reserve Bank of St. Louis. (n.d.). *Federal Reserve Economic Data (FRED)*. Retrieved September 2026, from <https://fred.stlouisfed.org>

Zillow Research. (n.d.). *Housing data*. Retrieved September 2026, from <https://www.zillow.com/research/>